

Math 526, S12

Quiz 11

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find the determinant of A where $A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix}$ using the co-factor expansion.

$$\text{Solution: } \det A = 1 \cdot \det \begin{bmatrix} 3 & 4 & 5 \\ 4 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix} - 2 \det \begin{bmatrix} 0 & 3 & 4 \\ 5 & 4 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

$$= \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix} - 2 \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix}$$

$$= (3 \cdot 0 - 4 \cdot 4) - 4(3 \cdot 0 - 4 \cdot 4)$$

$$= -16 - 4(-16)$$

$$= -16 + 64$$

$$= 48$$

Problem 2. (5 points) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Solve the linear system $Ax = b$ with

$$b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ by Cramer's Rule.}$$

$$\text{Solution: } \det A = -1 \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -1 \cdot (-1) + 3 \cdot (1) = 4.$$

$$B_1 = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \det B_1 = -1$$

$$B_2 = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix} \Rightarrow \det B_2 = 1$$

$$B_3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix} \Rightarrow \det B_3 = 1$$

$$\Rightarrow x_1 = \frac{|B_1|}{|A|} = -\frac{1}{4}, \quad x_2 = \frac{|B_2|}{|A|} = \frac{1}{4}, \quad x_3 = \frac{|B_3|}{|A|} = \frac{1}{4}.$$